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Full Length Article

SDCBNet: A sparse-to-dense cylinder-BEV fusion network for LiDAR semantic segmentation

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ABSTRACT

LiDAR semantic segmentation is a fundamental task in autonomous driving and robot perception. Although the existing voxel-based methods using cylindrical partition can adapt to the distribution characteristics of point clouds, they are limited by the modeling within a single cylinder and have difficulty effectively capturing the horizontal context information across cylinders, resulting in insufficient discriminative ability for sparse and minority classes. To address this limitation, we propose SDCBNet, a sparse-to-dense Cylinder-BEV network that transforms sparse cylindrical features into dense BEV representations. This method introduces a BEV branch on the basis of Cylinder3D and establishes an explicit injection mechanism between cylindrical features and the BEV space in multiple stages of encoding-decoding to enhance the modeling ability of horizontal context. Further, this paper designs the sparse-to-dense feature aggregation module SDFA, which generates a robust BEV representation through a dual-branch strategy of maximum pooling and attention aggregation; at the same time, it proposes the conditional routing feature selection module CRFS, which selectively injects cross features to suppress irrelevant context interference. Experimental results show that this method achieves 69.5% mIoU on the SemanticKITTI validation set, a 3.4% improvement compared to the Baseline, and achieves significant gains in multiple context-sensitive categories. Ablation experiments further verify the effectiveness of each proposed module.

1. Introduction

Understanding the surrounding scene is one of the most fundamental tasks for robots and autonomous vehicles. LiDAR semantic segmentation can provide fine-grained semantic understanding of the surrounding environment. Downstream tasks such as navigation, control and planning rely closely on the certainty of the perception method, so LiDAR semantic segmentation requires higher accuracy, consistency, and reliability. LiDAR provides long-range perception together with accurate geometric structure and reflection intensity measurements, making it a widely used sensor in autonomous perception systems.

Traditional point-based methods usually suffer from huge computational costs due to the processing dense point clouds, making it difficult to guarantee the real-time requirements of autonomous driving. Later, the voxel-based method was proposed, which quantizes the point cloud into voxel grids, significantly reducing the amount of data to be processed. Then, the sparse convolution method was proposed, which only performs 3D convolution operations on occupied voxels (Choy et al., 2019), enabling voxel methods to often achieve the most advanced performance. To overcome the high computational complexity caused by directly processing large-scale 3D point cloud data, researchers widely adopted projection-based segmentation strategies, typically projecting the point cloud into range images (Milioto et al., 2019) and Bird's-Eye Views (Zhang et al., 2020). These methods usually reduce the unordered 3D point cloud to a structured 2D representation, with the most common forms including range images generated through spherical projection and Bird's-Eye View (BEV). Thanks to the highly optimized 2D convolutional neural network (CNN) architecture, the projection-based method significantly reduces the computing time and fully ensures the real-time performance of the system. However, this dimensionality reduction projection process inevitably destroys the original geometric structure integrity of the point cloud. The generation process of range images is limited by the discretization of spherical grids, prone to introducing quantization errors and boundary blurring; moreover, the many-to-one mapping in the projection process can lead to spatial occlusion problems, resulting in the loss of key depth and topological geometric information, limiting the further improvement of segmentation accuracy. To solve this problem, multi-view fusion methods were proposed, fusing different views and finally performing feature fusion through

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point-level interaction, fully exploiting the complementary information of different views, making multi-view methods the mainstream of research.

Due to the significant sparsity and spatially uneven distribution characteristics of LiDAR point clouds, Cylinder3D (Zhu et al., 2020) relies on local feature aggregation within the cylinder to model the spatial structure. However, its modeling range is limited by the geometric boundary of a single cylinder, making it difficult to effectively capture horizontal spatial correlations across cylinders. For minority classes such as bicycle and person with smaller geometric sizes and highly dependent on context semantic discrimination, this local modeling approach is prone to insufficient discriminative evidence, causing their features to be submerged in the surrounding dominant categories (such as buildings, roads).

To alleviate these problems, this paper introduces the BEV stream based on the Cylinder3D baseline model and further proposes a multi-stage feature aggregation mechanism in the encoding-decoding architecture where the cylindrical features are explicitly fused into the BEV space in multiple intermediate layers. Unlike using BEV as a parallel branch or only using it in the output stage, we project the sparse cylindrical features onto the BEV plane at different scales, enabling the model to aggregate semantic information across cylinders in the XY plane, thereby constructing a more complete horizontal context representation. Furthermore, in the process of mapping the cylindrical features to the BEV, this paper designs a dual-branch sparse-to-dense aggregation strategy, combining attention-based scattering aggregation with maximum pooling, to retain the local significant geometric responses while highlighting the point features with semantic consistency within the same BEV unit. To avoid excessive interference from projected cylindrical features on the original BEV representations, we further introduce a gated residual fusion mechanism to selectively inject cylindrical information into the BEV feature space during the feature aggregation stage. Through the above multi-stage, cylinder to BEV feature injection design, the proposed method effectively compensates for the structural deficiencies of the cylindrical single representation in horizontal context modeling, and achieves significant performance improvements on multiple sparse and context-sensitive categories.

To further evaluate the generalization ability of our model beyond standard public benchmarks (e.g., SemanticKITTI (Behley et al., 2019) and nuScenes (Caesar et al., 2019)), we additionally collect a custom real-world campus dataset using a mobile platform equipped with a 16-beam LiDAR. This self-collected data presents a significantly sparser point distribution compared to standard benchmarks. By employing this data for zero-shot qualitative testing, we observe that SDCBNet exhibits favorable robustness against domain shifts. Specifically, the enhanced horizontal contextual awareness facilitated by our BEV stream effectively maintains structural continuity even in unconstrained, highly sparse physical environments, demonstrating its effectiveness in handling sparse real-world LiDAR scenarios.

We summarize the main contributions as:

- To address the limitations of cylindrical representations in contextual modeling, we propose a Cylinder–BEV dual-stream network with multi-stage Cylinder-to-BEV feature injection, which enhances global contextual perception.
- To enable robust Cylinder-to-BEV feature aggregation and adaptive feature injection under sparse and uneven point cloud distributions, we design two modules, Sparse-to-Dense Feature Aggregation (SDFA) and Conditional Routing Feature Selection (CRFS).
- To evaluate the effectiveness and generalization of our approach, we conduct experiments on SemanticKITTI and nuScenes, and further provide qualitative zero-shot evaluation on a self-collected sparse 16-beam campus dataset, suggesting its applicability to real-world LiDAR scenarios.

2. Related work

2.1. Voxel-based methods

Voxel-based methods discretize point clouds into sparse voxel grids, where multiple points may fall into the same voxel and are assigned a unified voxel label. 3D convolutions are then performed only on occupied voxels, which substantially improves computational efficiency. MinkowskiNet (Choy et al., 2019) proposed a general sparse convolution framework for spatio-temporal 3D point cloud data, modeling the time dimension as part of the spatial dimension. Cylinder3D (Zhu et al., 2020) increases the occupancy ratio of non-empty voxels, especially in long-range regions, through a cylindrical voxelization scheme and adopts asymmetric convolution kernels to capture anisotropic context along different axes, becoming a widely compared baseline method. NUC-Net (Wang et al., 2025) improved it by using non-equal division for the radial direction and using a larger resolution for different axial directions, more closely matching the dense and sparse distribution characteristics of the point cloud. SphereFormer (Lai et al., 2023) introduced slender windows in the radial direction, enabling a single attention calculation to aggregate context information over extremely long distances. AF^{2}S^{3}Net (Cheng et al., 2021) uses regular voxel grids to construct a multi-branch voxel extraction network, extracting features from different receptive fields and different levels, and explicitly modeling the importance relationship between different branch features to perform weighted fusion of features.

2.2. Projection-based methods

The projection-based method projects 3D point clouds onto structured 2D images, and then 2D convolutional neural networks (CNNs) are used for efficient semantic information extraction. These methods are mainly divided into two categories: Bird's-Eye View (BEV) and Range View. PolarNet (Zhang et al., 2020) proposed a BEV projection method based on polar coordinate representation, which captures global semantic information through radial and angular direction convolution operations. RangeNet++ (Milioto et al., 2019) uses a complete spherical projection to naturally adapt to the scanning structure of rotational LiDAR, and only downsamples along the horizontal direction while retaining height information. SalsaNext (Cortinhal et al., 2020) uses Residual Dilated Convolution Stack to significantly expand the receptive field without increasing the number of parameters, and replaces deconvolution with pixel-shuffle to reduce computational cost and avoid Checkerboard Artifacts. CENet (Cheng et al., 2022) handles range images with a lightweight backbone network and specific feature aggregation module at the minimum computational cost, overcoming the sparsity challenge in the projection process and achieving high-precision segmentation with extremely low latency. FRNet (Xu et al., 2023) regards the three-dimensional points sharing the same range pixel as a frustum, and retains and models all the point information within this frustum during the projection process, thus avoiding the omission of unprojected points.

2.3. Multi-view fusion methods

The point-based, voxel-based, and projection-based methods each have their own advantages and limitations. Integrating complementary information from different views is a reasonable approach to improving performance. The typical practice is to jointly model BEV and RV images (AMVNet, CPGNet, M2CNet) (Li et al., 2022; Liong et al., 2020; Zhou et al., 2026). AMVNet introduces a multi-view fusion mechanism driven by uncertainty modeling, explicitly estimating the confidence of the prediction results of points, voxels, and projection views, and performing adaptive weighting and refinement of multi-view outputs at the decision-making level. CPGNet proposes a cascaded network that realizes cross-view information transmission through bidirectional mapping of points and grids, and introduces consistency constraints to en-

hance the stability and semantic complementarity of cross-view fusion. RPVNet (Xu et al., 2021) constructs a multi-stage bidirectional feature transfer mechanism with points, voxels, and spherical projections as bridges to achieve deep interaction and complementary fusion between different views. PC-BEV (Qiu et al., 2025) integrates polar and Cartesian grid representations in the BEV space, operates on all positions within the BEV space, and achieves dense fusion while retaining more valuable contextual information.

3. Method

The proposed method framework is shown in Fig. 1. The original point cloud is processed by two parallel branches to extract complementary feature representations. In the Cylindrical Partition Branch, the input points are first encoded by a Point-wise MLP and discretized into cylindrical coordinates, and then input into a 3D sparse convolutional network to capture the voxel structure information. Meanwhile, the BEV Branch projects the point cloud onto the Bird's-Eye View grid through a max-pooling operation, generating a compact 2D representation. Finally, the extracted 3D voxel features and 2D BEV features are integrated through a Cylinder-to-BEV feature fusion module, and then concatenated to generate the final semantic segmentation result.

3.1. Cylinder and BEV backbone

Given a frame of LiDAR point cloud data, containing N points, each point is represented by its three-dimensional Cartesian coordinates (x, y, z) in the LiDAR coordinate system.

3.1.1. Cylinder stream

Due to the significant sparsity of the LiDAR point cloud in the radial direction and the continuous scanning characteristic in the azimuthal direction, we adopt the cylindrical coordinate system to reparameterize the point cloud to enhance the consistency of spatial structure expression. Specifically, each point (x, y, z) is mapped to cylindrical coordinates (ρ, θ, z), defined as follows:

$$\begin{pmatrix} \rho \\ \theta \\ z \end{pmatrix} = \begin{pmatrix} \sqrt{x^2 + y^2} \\ \arctan2(y, x) \\ z \end{pmatrix} \tag{1}$$

Here, ρ represents the radial distance and θ represents the azimuth angle. This representation helps alleviate the uneven distribution problem caused by the density difference between distant and nearby points, and maintains the continuity of the angular dimension. However, this mechanism is inherently limited by the geometric boundaries of a single cylinder: on the one hand, the division of cylinders on the horizontal plane separates adjacent areas, making it difficult to effectively establish long-range semantic dependencies across cylinders in the early stage; Based on this, we introduce the BEV stream and explicitly project and fuse the cylindrical feature to the BEV space at multiple scale levels of encoding-decoding, enabling the semantic across cylinders to gradually propagate and aggregate at different resolutions, forming a more complete horizontal context expression.

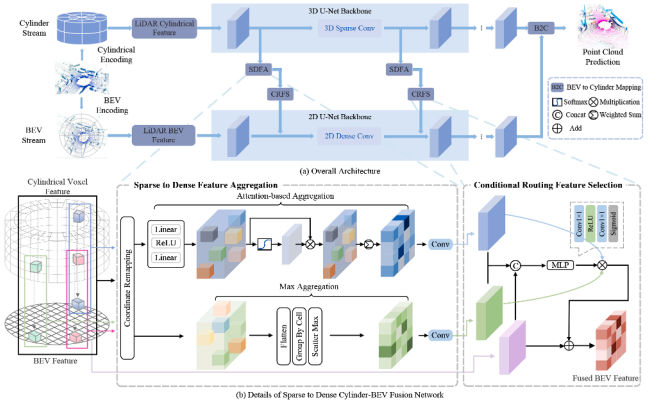


Fig. 1. (a) The framework of our method. We proposed a dual-stream network. Firstly, voxel features and BEV features were obtained through Cylindrical voxel encoding and BEV encoding respectively, and were sent into the sparse convolutional network and the two-dimensional convolutional network respectively. During different encoding and decoding stages of the network, multi-scale cylinder-to-BEV feature injection and fusion were carried out to predict the results. (b) The details represented by SDCB. The Cylindrical Voxel features were first mapped to Cartesian coordinates through coordinate remapping, then mapped to the BEV grid through the dual-path feature aggregation method, and finally, adaptive feature selection was performed within the grid by the CRFS module.

3.1.2. BEV stream

To address the insufficient cross-column correlation of the cylindrical voxels on the horizontal plane and the long-tail distribution problem caused by distant sparse points, we reorganized the point cloud in the Bird's-Eye View (BEV) into a regular two-dimensional grid representation. For a given point (x, y, z), it is first restricted within the preset spatial range and uniformly quantized according to the resolution (W, H):

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \frac{x - x_{\min}}{x_{\max} - x_{\min}} \cdot W_{bev} \\ \frac{y - y_{\min}}{y_{\max} - y_{\min}} \cdot H_{bev} \end{pmatrix} \tag{2}$$

Here, (u, v) represents the discrete index of the point in the BEV grid. For each grid cell, we encode 7-dimensional point features$[x, y, z, r, d, \Delta x, \Delta y]$ to enhance local position perception and improve perception of the local geometric structure, where r denotes the reflection intensity, d represents the radial distance from the LiDAR sensor, and Δx, Δy denote the relative offsets to the center of the corresponding BEV grid cell. Subsequently, we will aggregate the sparse features falling within each BEV cell to form a dense BEV feature map, providing structured input for subsequent two-dimensional convolution encoding and cross-view fusion.

3.2. Sparse to dense feature aggregation

To explicitly aggregate the horizontal context information across cylinders in the BEV plane, we propose a projection fusion module named Sparse to Dense Feature Aggregation, which projects from the sparse cylindrical voxel features to the dense BEV feature maps. This module bridges the domain gap between the sparse cylindrical voxel features (in polar coordinates) and the dense BEV features (in Cartesian coordinates) through geometrically aligned projection transformations and a dual-branch feature aggregation mechanism.

3.2.1. Cylinder to BEV coordinate mapping

To align the sparse features of the cylindrical branches onto the BEV plane, we establish a geometric mapping from the cylindrical grid indices to the BEV grid cells. Specifically, given the radial and azimuthal discrete indices (r, θ) of a cylindrical voxel, we first restore it to the physical polar coordinates (ρ, θ) corresponding to the grid center, then convert it to the Cartesian coordinates (x, y) on the horizontal plane, and finally quantize it to the BEV grid index (u, v) according to the spatial range and resolution of the BEV. This mapping explicitly projects the local representation of the cylindrical space onto a regular two-dimensional grid, which is used for subsequent semantic aggregation across cylinders in

the BEV plane.

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \left\lfloor \frac{\rho\cos\theta - x_{\min}}{x_{\max} - x_{\min}} \cdot W \right\rfloor \\ \left\lfloor \frac{\rho\sin\theta - y_{\min}}{y_{\max} - y_{\min}} \cdot H \right\rfloor \end{pmatrix} \tag{3}$$

Among them, ρ and θ represent the radial distance and azimuth angle of the center of the cylindrical grid cell, respectively. W and H are the resolution of the BEV feature map in the horizontal and vertical directions, respectively. The obtained (u, v) represents continuous positions, and in the subsequent implementation, discrete BEV pixel indices are obtained through rounding and boundary clipping.

3.2.2. Dual branch aggregation

Due to the inherent non-bijective mapping relationship between the cylindrical coordinate system and the Cartesian coordinate system, this process essentially constitutes a many-to-one projection from sparse 3D voxels to dense 2D grids. To address this feature conflict and retain complete information, we propose Dual Branch Aggregation.

The Max Pooling Branch is used to retain the most discriminative responses within each BEV unit, being more robust to noise and redundant points, and can naturally highlight significant geometric and semantic cues:

$$F_i^{max} = \max_{p \in \mathcal{P}_i} f_p \tag{4}$$

Among them, $\mathcal{P}_i$ represents the set of all points mapped to the ith BEV unit, and f_p is the feature of point p. The maximum pooling can effectively suppress noise and redundancy, maintain numerical stability in scenarios with significant changes in point density, and provide a structural inductive bias for the model.

To capture more comprehensive context information and overcome the limitations of maximum pooling, we introduce a soft aggregation path, achieving weighted aggregation normalized by the BEV unit through point-level attention. Specifically, we use a hierarchical-related lightweight scoring network MLP to evaluate the importance of each point, and then perform Softmax normalization (cell-wise softmax) within the current BEV unit to obtain the weight α_p:

$$\alpha_p = \mathrm{softmax}(\mathrm{MLP}(f_p)) \tag{5}$$

Subsequently, the point features within the unit are weighted and summed to form a consistency feature representation F_i^{attn}:

$$F_i^{attn} = \sum_{p \in \mathcal{P}_i} \alpha_p f_p \tag{6}$$

This cell-wise attention-based aggregation method can adaptively smooth out outliers and noise based on local context, while maintaining the overall consistency of feature distribution. With dual-branch aggregation design, the model can simultaneously capture the most discriminative local significant features and the smooth context reflecting the overall distribution within each BEV unit. The two complement each other in feature representation, laying a good foundation for the dynamic routing and fusion of multiple source features in the subsequent stage.

3.3. Conditional routing feature selection

Although the dual-branch aggregation in SDFA can provide complementary contextual information for each BEV grid, directly re-feeding these aggregated features into the BEV backbone is not the optimal choice. This is because, in the features obtained from the cylindrical branch projection, there often simultaneously contain beneficial cross-cylindrical contextual information as well as unreliable responses introduced by the many-to-one projection process. Therefore, a uniform direct fusion strategy is prone to inject irrelevant or even noise information into the already relatively reliable BEV representation, thereby weakening the class discrimination ability rather than improving it. To solve this problem, we further introduce the Conditional Routing Feature Selection (CRFS) module, which performs content-aware selective injection of cylindrical features before they are fed back to the BEV branches. Specifically, let F_{bev} represent the current BEV backbone features, F_{attn} and $F_{\max}$ represent the two types of cylindrical projection features obtained from the attention aggregation branch and the maximum pooling branch respectively. Unlike directly treating these features equally, we first concatenate the three of them in the channel dimension to construct a context-aware routing input:

$$F_{cat} = \mathrm{Conv}\big(\mathrm{Concat}(F_{bev}, F_{attn}, F_{max})\big) \tag{7}$$

This design enables the gated decision to not only explicitly depend on the query state of the current BEV unit (query), but also be able to perceive the contextual evidence aggregated by the cylindrical projection, thereby achieving input-dependent feature modulation. Based on this joint representation, a lightweight nonlinear gated function is used to predict the element-wise routing weights corresponding to the two auxiliary branches:

$$[G_{attn}, G_{max}] = \sigma(\phi(F_{cat})) \tag{8}$$

Among them, $\phi(\cdot)$ represents the learnable transformation function, and $\sigma(\cdot)$ represents the Sigmoid activation function. In this way, the model can adaptively determine how much information should be retained in each projection branch based on the semantic state of the current BEV grid. Under different BEV semantic conditions, the model can adaptively suppress irrelevant or noisy context responses while retaining discriminative geometric evidence. Ultimately, the cylindrical features that have undergone gating modulation are injected into the BEV backbone representation in the form of residuals:

$$F'_{bev} = F_{bev} + G_{attn} \odot F_{attn} + G_{max} \odot F_{max} \tag{9}$$

where $\odot$ denotes element-wise multiplication. This residual design preserves the stability of the original BEV representation while enabling discriminative cross-cylinder context to be selectively integrated. Overall, CRFS enables the network to inject auxiliary cross-cylinder context back into the BEV branch in a controlled and discriminative manner, thereby effectively strengthening the modeling capability for weak categories and context-sensitive categories without interfering with dominant class representations.

4. Experiments

This section presents the experimental evaluation of the proposed model. We conduct comprehensive experiments on two large-scale public benchmarks, SemanticKITTI (Behley et al., 2019) and nuScenes (Caesar et al., 2019), comparing our method against representative point-based, projection-based, voxel-based, and multi-view fusion architectures. To further validate the model's generalization capability, we provide additional evaluations on a custom real-world dataset. The remainder of this section is organized as follows: Sections 4.1 and 4.2 introduce the datasets and implementation details. Section 4.3 reports the qualitative, quantitative, and distance-based results on SemanticKITTI. Section 4.4 presents the results on nuScenes. Section 4.5 provides the ablation studies, and Section 4.6 evaluates the model on the self-collected real-world dataset.

4.1. Public dataset

4.1.1. SemanticKITTI

SemanticKITTI (Behley et al., 2019) is a large-scale dataset collected via a Velodyne HDL-64E sensor. Following the official guidelines, we use sequences 00–07 and 09–10 for training, sequence 08 for validation, and sequences 11–21 for testing. Although the raw dataset provides annotations for 28 classes, the labels are mapped to 19 valid categories for training and evaluation by merging rare classes and distinguishing moving objects, adhering to the standard benchmark protocol.

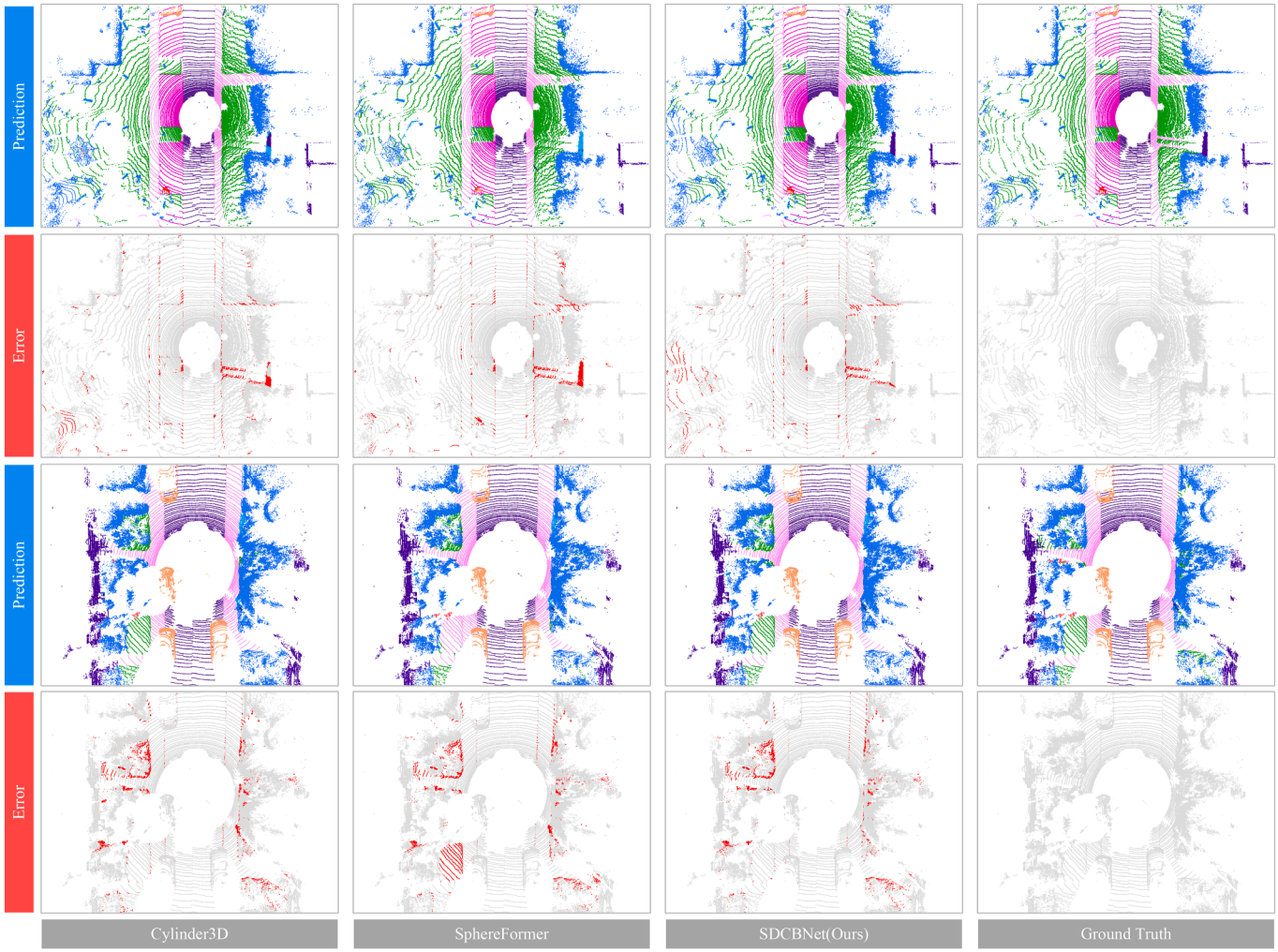


Fig. 2. Qualitative analysis of different methods on the SemanticKITTI 08 sequence. The first and third rows represent the predictions of different methods, while the second and fourth rows represent the differences between the corresponding predictions and the ground truth.

4.1.2. NuScenes

The nuScenes dataset (Caesar et al., 2019) provides large-scale LiDAR point clouds captured by a Velodyne HDL-32E sensor. Following the official guidelines, the 1000 driving scenes are partitioned into 700, 150, and 150 scenes for training, validation, and testing, respectively. While the original point clouds are annotated with 32 semantic classes, these are mapped to 16 official categories for learning and evaluation in the LiDAR semantic segmentation task (see Table 2).

4.2. Implementation details

We implement and train our proposed SDCBNet using two NVIDIA A30 GPUs. The input point clouds are transformed into cylindrical voxel and Bird's-Eye-View (BEV) representations for subsequent feature extraction. During the 50-epoch training process, we apply data augmentation and set the per-GPU batch size to 2. The network is optimized using SGD with Nesterov momentum (0.9) and a weight decay of 1e-4. The initial learning rate is set to 0.01, modulated by a cosine decay schedule following a 2-epoch linear warmup.

To measure the disparity between the predicted segmentation results and the ground truth labels, we employ the mean Intersection over Union (mIoU) to evaluate the model's overall performance. The IoU for a specific class c is calculated as follows:

$$IoU_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{10}$$

$$mIoU = \frac{1}{C}\sum_{c=1}^{C} IoU_c \tag{11}$$

where C denotes the total number of classes, while TP_c, FP_c, and FN_c represent the number of true positives, false positives, and false negatives for class c, respectively.

4.3. Evaluated on SemanticKITTI

4.3.1. Qualitative results

To provide an intuitive understanding of our method's performance, Fig. 2 visualizes the qualitative results on representative SemanticKITTI scenes. Compared to Cylinder3D and SphereFormer, SDCBNet not only yields more coherent boundaries but also better maintains the geometric integrity of small objects in locally sparse or occluded areas. Furthermore, our method demonstrates significant advantages when handling large-scale features. By accurately capturing the global geometric shape of large planar structures, SDCBNet successfully recovers a more accurate and coherent overall morphology for vegetation and buildings.

4.3.2. Quantitative results

Consistent with the qualitative observations, the quantitative results on the SemanticKITTI validation set further demonstrate the effectiveness of the proposed method. As shown in Table 1, SDCBNet achieves an mIoU of 69.5%, yielding a 3.4% improvement over the baseline.

Table 1
Segmentation mIoU (%) results on the SemanticKITTI validation dataset.

Method	mIoU	car	bicycle	motorcycle	truck	other-vehicle	person	bicyclist	motorcyclist	road	parking	sidewalk	other-ground	building	fence	vegetation	trunk	terrain	pole	traffic-sign
RangeNet++ (Milioto et al., 2019)	52.8	91.0	25.0	47.1	40.7	25.5	45.2	62.9	0.0	93.8	46.5	81.9	0.2	85.8	54.2	84.2	52.9	72.7	53.2	40.0
PolarNet (Zhang et al., 2020)	53.6	91.5	30.7	38.8	46.4	24.0	54.1	62.2	0.0	92.4	47.1	78.0	**1.8**	89.1	45.5	85.4	59.6	72.3	58.1	42.2
GFNet (Qiu et al., 2022)	63.0	94.2	49.7	63.2	74.9	32.1	69.3	83.2	0.0	**95.7**	53.8	83.8	0.2	**91.2**	**62.9**	**88.5**	66.1	**76.2**	64.1	48.3
AMVNet (Liong et al., 2020)	65.2	95.6	48.8	65.4	**88.7**	54.8	70.8	86.2	0.0	95.5	53.9	83.2	0.3	90.9	62.1	87.9	66.8	74.2	64.7	49.3
PC-BEV (Qiu et al., 2025)	66.4	94.3	**67.6**	71.4	75.4	67.5	77.4	81.3	0.1	95.2	39.7	83.0	0.1	86.3	49.7	85.1	**70.2**	68.2	**70.2**	**64.2**
Baseline	66.1	**96.9**	54.4	76.0	81.0	67.0	79.3	92.4	0.1	94.5	47.9	82.2	0.1	90.3	56.9	85.9	66.9	69.2	63.6	50.6
Ours	**69.5**	**96.9**	59.8	**85.4**	88.0	**69.4**	**81.8**	**93.9**	**14.5**	95.1	**55.1**	**83.9**	0.0	91.0	61.5	87.7	70.0	73.0	66.5	47.9

Table 2
Segmentation mIoU (%) results on the SemanticKITTI test dataset.

	Method	mIoU	car	bicycle	motorcycle	truck	other-vehicle	person	bicyclist	motorcyclist	road	parking	sidewalk	other-ground	building	fence	vegetation	trunk	terrain	pole	traffic-sign
Point	PointNet (Qi et al., 2016)	14.6	46.3	1.3	0.3	0.1	0.8	0.2	0.2	0.0	61.6	15.8	35.7	1.4	41.4	12.9	31.0	4.6	17.6	2.4	3.7
	PointNet++ (Qi et al., 2017)	20.1	53.7	1.9	0.2	0.9	0.2	0.9	1.0	0.0	72.0	18.7	41.8	5.6	62.3	16.9	46.5	13.8	30.0	6.0	8.9
	RandLA-Net (Hu et al., 2019)	53.9	94.2	26.0	25.8	40.1	38.9	49.2	48.2	7.2	90.7	60.3	73.7	20.4	86.9	56.3	81.4	61.3	66.8	49.2	47.7
	KPConv (Thomas et al., 2019)	58.8	96.0	30.2	42.5	33.4	44.3	61.5	61.6	11.8	88.8	61.3	72.7	31.6	90.5	64.2	84.8	69.2	69.1	56.4	47.4
Voxel	PCSCNet (Park et al., 2022)	62.7	95.7	48.8	46.2	36.4	40.6	55.5	68.4	**55.9**	89.1	60.2	72.4	23.7	89.3	64.3	84.2	68.2	68.1	60.5	63.9
	JS3C-Net (Yan et al., 2020)	66.0	95.8	59.3	52.9	54.3	46.0	69.5	65.4	39.9	88.9	61.9	72.1	31.9	92.5	70.8	84.5	69.8	67.9	60.7	**68.7**
	Cylinder3D (Zhu et al., 2020)	67.8	97.1	**67.6**	**64.0**	**59.0**	**58.6**	73.9	67.9	36.0	91.4	65.1	75.5	32.3	91.0	66.5	85.4	71.8	68.5	62.6	65.6
Single-view	RangeNet++ (Milioto et al., 2019)	52.2	91.4	25.7	34.4	25.7	23.0	38.3	38.8	4.8	91.8	65.0	75.2	27.8	87.4	58.6	80.5	55.1	64.6	20.2	26.3
	PolarNet (Zhang et al., 2020)	54.3	93.8	40.3	30.1	22.9	28.5	43.2	40.2	5.6	90.8	61.7	74.4	21.7	90.0	61.3	84.0	65.5	67.8	51.8	57.5
	SalsaNext (Cortinhal et al., 2020)	59.5	91.9	48.3	38.6	38.9	31.9	60.2	59.0	19.4	91.7	63.7	75.8	29.1	90.2	64.2	81.8	63.6	66.5	54.3	62.1
	SwinURNet (Wang et al., 2024)	61.6	93.0	45.9	45.1	45.2	42.3	63.7	63.0	27.1	90.8	66.4	74.7	28.6	90.5	64.1	82.2	65.0	66.4	56.3	61.2
	RangeViT (Ando et al., 2023)	64.0	95.4	55.8	43.5	29.8	42.1	63.9	58.2	38.1	93.1	70.2	80.0	32.5	92.0	69.0	85.3	70.6	71.2	60.8	64.7
	LENet (Ding, 2023)	64.5	93.9	57.0	51.3	44.3	44.4	66.6	64.9	36.0	91.8	68.3	76.9	30.5	91.2	66.0	83.7	68.3	67.8	58.6	63.2
Multi-view	MPF (Alnaggar et al., 2020)	55.5	93.4	30.2	38.3	26.1	28.5	48.1	46.1	18.1	90.6	62.3	74.5	30.6	88.5	59.7	83.5	59.7	69.2	49.7	58.1
	AMVNet (Liong et al., 2020)	65.3	96.2	59.9	54.2	48.8	45.7	71.0	65.7	11.0	90.1	71.0	75.8	32.4	92.4	69.1	85.6	71.7	69.6	62.7	67.2
	GFNet (Qiu et al., 2022)	65.4	96.0	53.2	48.3	31.7	47.3	62.8	57.3	44.7	93.6	72.5	**80.8**	31.2	94.0	**73.9**	85.2	71.1	69.3	61.8	68.0
	M2CNet (Zhou et al., 2026)	65.7	96.5	55.3	52.9	32.1	48.4	60.8	59.5	43.2	**93.9**	72.8	76.0	**32.9**	**94.7**	71.2	**86.7**	72.6	**71.5**	62.8	65.4
	SPVNAS (Tang et al., 2020)	67.0	**97.2**	50.6	50.4	56.6	58.0	67.4	67.1	50.3	90.2	67.6	75.4	21.8	91.6	66.9	86.1	**73.4**	71.0	64.3	67.3
	PC-BEV (Qiu et al., 2025)	67.2	96.5	66.5	57.7	37.6	48.2	62.5	68.0	49.1	93.0	**73.5**	79.8	32.1	92.1	69.2	85.4	70.0	70.0	62.4	63.0
	Ours	**67.9**	96.8	66.6	60.2	49.3	54.3	**74.7**	**75.1**	23.8	91.9	70.3	77.5	30.5	91.6	68.4	85.9	71.8	69.8	**65.0**	67.4

Table 3
Distance-based evaluation on the SemanticKITTI validation set. Overall Accuracy (Acc) and Intersection over Union (IoU) are reported in percentage (%). Class-specific results (Car, Person, Road) represent their respective IoU scores.

Method	0-10m					10-20m					20-30m					30-40m					40-50m				
	Acc	mIoU	Car	Person	Road	Acc	mIoU	Car	Person	Road	Acc	mIoU	Car	Person	Road	Acc	mIoU	Car	Person	Road	Acc	mIoU	Car	Person	Road
SalsaNext	89.6	58.5	90.4	73.6	95.4	86.6	54.6	80.3	58.2	94.1	83.2	44.5	68.9	40.3	86.7	81.2	36.9	54.0	17.5	80.6	80.7	31.2	41.6	14.5	76.0
Cylinder3D	91.8	66.8	**97.5**	86.6	95.1	91.2	65.6	96.8	**76.8**	94.3	89.8	60.5	93.9	68.1	91.0	88.6	50.3	84.0	49.3	86.5	88.3	42.5	74.2	**25.5**	80.3
SphereFormer	92.8	68.1	97.2	85.2	95.2	**92.3**	67.0	96.6	71.9	94.4	**90.9**	63.3	94.6	66.0	91.7	**90.2**	**56.7**	**88.2**	**56.4**	87.8	**89.9**	**47.7**	**79.9**	40.0	83.2
SDCBNet (Ours)	**93.0**	**70.3**	97.3	**91.5**	**95.6**	91.9	**69.4**	**97.3**	**76.8**	**94.7**	90.5	**64.0**	**95.1**	**68.7**	**91.9**	89.5	51.5	86.9	45.0	**88.0**	89.3	42.9	78.5	24.6	**84.0**

Table 4
Segmentation mIoU (%) results on the nuScenes validation set.

	Method	mIoU	barrier	bicycle	bus	car	construction	motorcycle	pedestrian	traffic-cone	trailer	truck	driveable	other	sidewalk	terrain	manmade	vegetation
Single-view	RangeNet++ (Milioto et al., 2019)	65.5	66.0	21.3	77.2	80.9	30.2	66.8	69.6	52.1	54.2	72.3	94.1	66.6	63.5	70.1	83.1	79.8
	PolarNet (Zhang et al., 2020)	71.0	74.7	28.2	85.3	90.9	35.1	77.5	71.3	58.8	57.4	76.1	96.5	71.1	74.7	74.0	87.3	85.7
	SalsaNext (Cortinhal et al., 2020)	72.2	74.8	34.1	85.9	88.4	42.2	72.4	72.2	63.1	61.3	76.5	96.0	70.8	71.2	71.5	86.7	84.4
	RangeViT (Ando et al., 2023)	75.2	75.5	40.7	88.3	90.1	49.3	79.3	77.2	66.3	65.2	80.0	96.4	71.4	73.8	73.8	89.9	87.2
Voxel	PCSCNet (Park et al., 2022)	72.0	73.3	42.2	87.8	86.1	44.9	82.2	76.1	62.9	49.3	77.3	95.2	66.9	69.5	72.3	83.7	82.5
	Cylinder3D (Zhu et al., 2020)	76.1	76.4	40.3	91.2	93.8	51.3	78.0	78.9	64.9	62.1	84.4	96.8	71.6	76.4	75.4	90.5	87.4
	SphereFormer (Lai et al., 2023)	78.4	77.7	43.8	94.5	93.1	52.4	86.9	**81.2**	65.4	73.4	85.3	97.0	73.4	75.4	75.0	**91.0**	**89.2**
Multi-view	AMVNet (Liong et al., 2020)	76.1	79.8	32.4	82.2	86.4	**62.5**	81.9	75.3	**72.3**	**83.5**	65.1	**97.4**	67.0	**78.8**	74.6	90.8	87.9
	RPVNet (Xu et al., 2021)	77.6	78.2	43.4	92.7	93.2	49.0	85.7	80.5	66.0	66.9	84.0	96.9	73.5	75.9	70.6	90.6	88.9
	PC-BEV (Qiu et al., 2025)	**78.8**	78.2	**46.3**	92.5	93.4	55.0	**87.1**	81.0	65.4	69.2	**85.7**	97.1	**76.8**	77.0	76.3	90.6	88.5
	SDCBNet(Ours)	77.5	**85.6**	43.4	**95.7**	**94.1**	52.2	80.1	77.8	68.3	64.4	83.1	97.2	68.7	74.5	**78.7**	88.9	88.1

The most pronounced gains are observed for small and sparsely sampled foreground categories, including motorcycle, person, bicyclist, and motorcyclist.

These categories usually contain only a limited number of LiDAR points and occupy few cylindrical voxels. After cylindrical partitioning, their already sparse observations may be further distributed across neighboring cylinder cells, resulting in fragmented local representations and insufficient discriminative evidence within each individual cylinder. Moreover, weak foreground responses can easily be overwhelmed by dominant surrounding categories such as road, building, and vegetation.

SDCBNet alleviates this problem by introducing a BEV stream that aggregates horizontal context across neighboring cylindrical partitions. The max-aggregation branch preserves salient foreground responses and prevents sparse but discriminative features from being diluted, while the attention-based branch emphasizes semantically consistent evidence within each BEV cell. CRFS further evaluates the current BEV feature together with the two aggregated features and selectively injects reliable cross-view information while suppressing noisy responses caused by sparse observations and many-to-one projection. Through multi-stage feature injection at different resolutions, the model progressively complements fragmented local evidence with broader contextual information, leading to more complete representations for small and sparse foreground objects.

The improvements on large structured regions, such as parking and sidewalk, further indicate that the proposed Cylinder-BEV interaction is also effective in modeling broad horizontal dependencies and preserving spatial continuity.

4.3.3. Distance-based evaluation results

To further investigate the robustness of the proposed model under varying spatial distributions, we conduct a distance-based evaluation on the SemanticKITTI validation set. The point clouds are partitioned into five distance intervals, i.e., 0–10 m, 10–20 m, 20–30 m, 30–40 m, and 40–50 m, according to their distance from the LiDAR sensor. The corresponding mIoU and class-specific IoU results are reported in Table 3, while the overall performance trends are illustrated in Fig. 3.

As the sensing distance increases, all methods experience performance degradation because distant objects are represented by fewer and more irregularly distributed points. Nevertheless, SDCBNet achieves the highest mIoU in the 0–10 m, 10–20 m, and 20–30 m intervals, with scores of 70.3%, 69.4%, and 64.0%, respectively. This advantage can be attributed to the complementary roles of the cylindrical and BEV representations. The cylindrical stream preserves local geometric details, whereas the BEV stream aggregates horizontal context across neighboring cylindrical partitions. Their multi-stage interaction therefore provides more complete structural evidence in the near-to-middle ranges, where sufficient local observations remain available for cross-view fusion.

At longer distances, SphereFormer achieves higher overall mIoU than SDCBNet. Its spherical-window attention effectively captures long-range dependencies along the radial direction and aggregates contextual information in extremely sparse regions, leading to stronger overall performance in the 30–40 m and 40–50 m intervals.

Despite this, SDCBNet retains the highest IoU for the Road category across almost all distance intervals, achieving 88.0% at 30–40 m and 84.0% at 40–50 m. Road regions usually form large and continuous surfaces in the BEV plane, whereas their cylindrical features become increasingly fragmented as the point density decreases. CRFS addresses this issue by jointly considering the current BEV feature and the max- and attention-aggregated features to predict content-dependent routing weights. It selectively incorporates reliable cross-view information while suppressing noisy responses, thereby preserving the spatial continuity and semantic consistency of road regions under sparse observations.

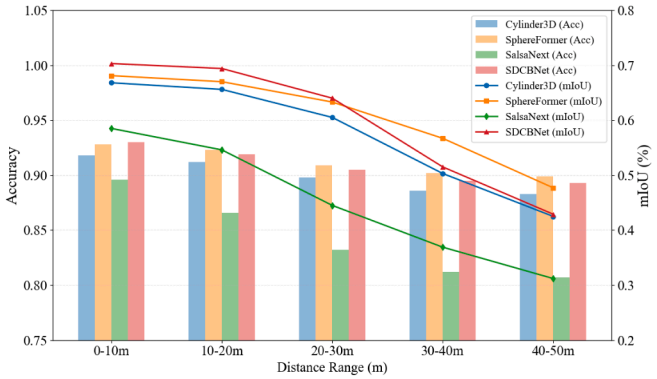


Fig. 3. Distance-based segmentation performance comparison on the SemanticKITTI validation set. The performance is evaluated in terms of mIoU and accuracy.

Overall, SphereFormer shows stronger overall performance in extremely sparse long-range regions, while SDCBNet is more effective in preserving large-scale continuous structures through BEV modeling and conditional cross-view routing. These results demonstrate the complementary strengths of the two architectures across different distance ranges.

4.4. Evaluation on nuScenes validation set

In addition to SemanticKITTI, we further evaluate SDCBNet on the nuScenes dataset to verify its generalization ability in more diverse urban driving environments. Although SemanticKITTI has already demonstrated the effectiveness of our method, nuScenes differs in scene composition, category taxonomy, and object distribution. Therefore, evaluation on nuScenes provides a complementary benchmark for assessing the robustness and transferability of SDCBNet across datasets.

Table 4 presents the quantitative semantic segmentation results on the nuScenes validation set. Our SDCBNet achieves an mIoU of 77.5%, outperforming several baseline methods including RangeNet++, PolarNet, and AMVNet.

While PC-BEV and RPVNet achieve higher overall mIoU scores, a detailed class-wise analysis reveals the specific characteristics of our approach in parsing urban environments. First, it shows strong performance in segmenting dynamic moving vehicles, particularly on categories such as bus, car, and bicycle. Second, the model demonstrates robustness in identifying structural obstacles and traffic infrastructure, including barrier and traffic-cone. Finally, it is effective in delineating continuous ground geometries, such as terrain and driveable surface. This fine-grained segmentation performance indicates that the proposed Sparse-to-Dense Feature Aggregation module can capture contextual features while preserving spatial details.

4.5. Ablation studies

To evaluate the contribution of each proposed component, we conduct ablation studies on the SemanticKITTI validation set. All variants are trained under the same experimental settings for fair comparison, and the detailed class-wise results are reported in Table 5.

4.5.1. Effectiveness of sparse-to-dense feature aggregation (SDFA)

The SDFA module is designed to aggregate point-level semantic cues into the BEV representation. To analyze its effect, we evaluate the attention-based branch and the max-pooling branch individually and jointly (Methods II, III, and IV). As shown in Table 5, adding the attention-based aggregation (Method II) improves the baseline mIoU from 66.1% to 67.5%, indicating that semantic-aware aggregation is

Table 5
Ablation study of different components on SemanticKITTI validation set. All values are IoU (%).

Method	Base	Attn	Max	CRFS	mIoU	car	bicycle	motorcycle	truck	other-vehicle	person	bicyclist	motorcyclist	road	parking	sidewalk	other-ground	building	fence	vegetation	trunk	terrain	pole	traffic-sign
I	✓				66.1	96.9	54.4	76.0	81.0	67.0	79.3	92.4	0.1	94.5	47.9	82.2	0.1	90.3	56.9	85.9	66.9	69.2	63.6	50.6
II	✓	✓			67.5	96.5	59.5	81.0	83.4	65.6	77.5	91.7	10.2	94.9	50.5	83.1	0.0	90.7	58.0	86.9	66.9	71.5	64.0	**50.7**
III	✓		✓		68.1	96.4	55.5	81.3	82.0	63.0	77.0	91.5	**32.1**	94.5	46.8	82.7	0.1	90.3	57.6	87.5	67.6	72.7	64.5	50.3
IV	✓	✓	✓		68.2	96.4	57.1	80.4	80.1	64.0	78.2	93.4	19.1	94.8	48.6	82.9	**0.2**	**91.0**	60.3	**88.7**	68.4	**76.1**	65.6	49.7
V	✓	✓	✓	✓	**69.5**	**96.9**	**59.8**	**85.4**	**88.0**	**69.4**	**81.8**	**93.9**	14.5	**95.1**	**55.1**	**83.9**	0.0	**91.0**	**61.5**	87.7	**70.0**	73.0	**66.5**	47.9

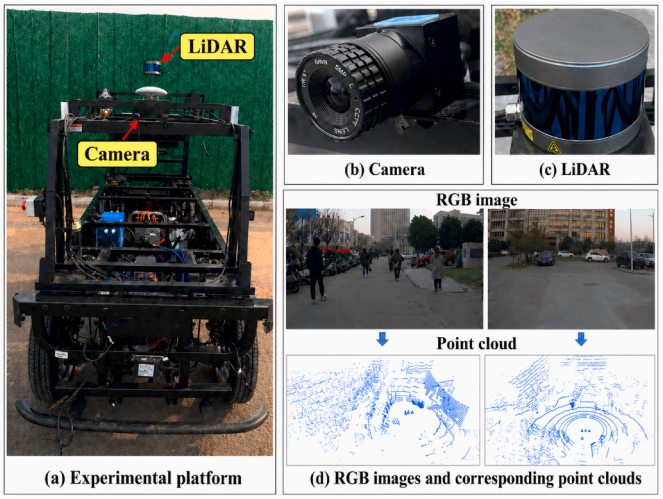


Fig. 4. Experimental platform and data samples. (a) The data acquisition platform equipped with a camera and a LiDAR. (b) The onboard camera. (c) The LiDAR sensor (Velodyne VLP-16). (d) Sample RGB images of the campus environment and their corresponding LiDAR point clouds.

beneficial for BEV feature enhancement. Replacing it with the max-aggregation branch (Method III) further increases the mIoU to 68.1%, and yields a particularly notable improvement on the motorcyclist category, which rises from 0.1% to 32.1%. When the two branches are combined (Method IV), the mIoU is further improved to 68.2%, with additional gains on several large-area background categories such as terrain and vegetation. However, the limited improvement over Method III suggests that directly combining the two branches is not sufficient to fully reconcile their heterogeneous feature responses.

4.5.2. *Impact of conditional routing feature selection (CRFS)*

Based on the dual-branch SDFA design, we further introduce the CRFS module to adaptively regulate the contribution of multi-stream features according to the current BEV semantic state. As shown in Table 5, CRFS (Method V) further improves the overall mIoU from 68.2% to 69.5%, achieving the best overall performance among all ablated variants.

In terms of class-wise results, CRFS brings clear improvements on several challenging foreground categories, including motorcycle, truck, person, and bicyclist, while also enhancing structural categories such as fence, trunk, and pole. These results indicate that CRFS helps produce a better balance between local detail preservation and contextual feature selection. Although a few categories still exhibit fluctuations, the consistent gain in overall mIoU demonstrates the effectiveness and robustness of the proposed routing mechanism.

4.6. *Evaluated on real-world dataset*

To rigorously evaluate the practical adaptability and zero-shot generalization of our proposed model, we constructed an independent, real-world test set. As illustrated in Fig. 4, data acquisition was conducted using a mobile platform equipped with a Velodyne VLP-16 LiDAR alongside 6mm and 12mm cameras. In total, we collected 820 frames of LiDAR point clouds capturing various campus road scenarios.

Unlike the high-density data often found in standard commercial datasets, our self-collected data is generated by 16-beam LiDAR scans, which inherently produce sparser and more irregular point distributions typical of low-cost autonomous driving setups. It is important to emphasize that this custom dataset was strictly excluded from the model training phase and was reserved entirely for unseen testing.

As shown in Fig. 5, our model demonstrates strong generalization capabilities in unconstrained physical environments. Despite the

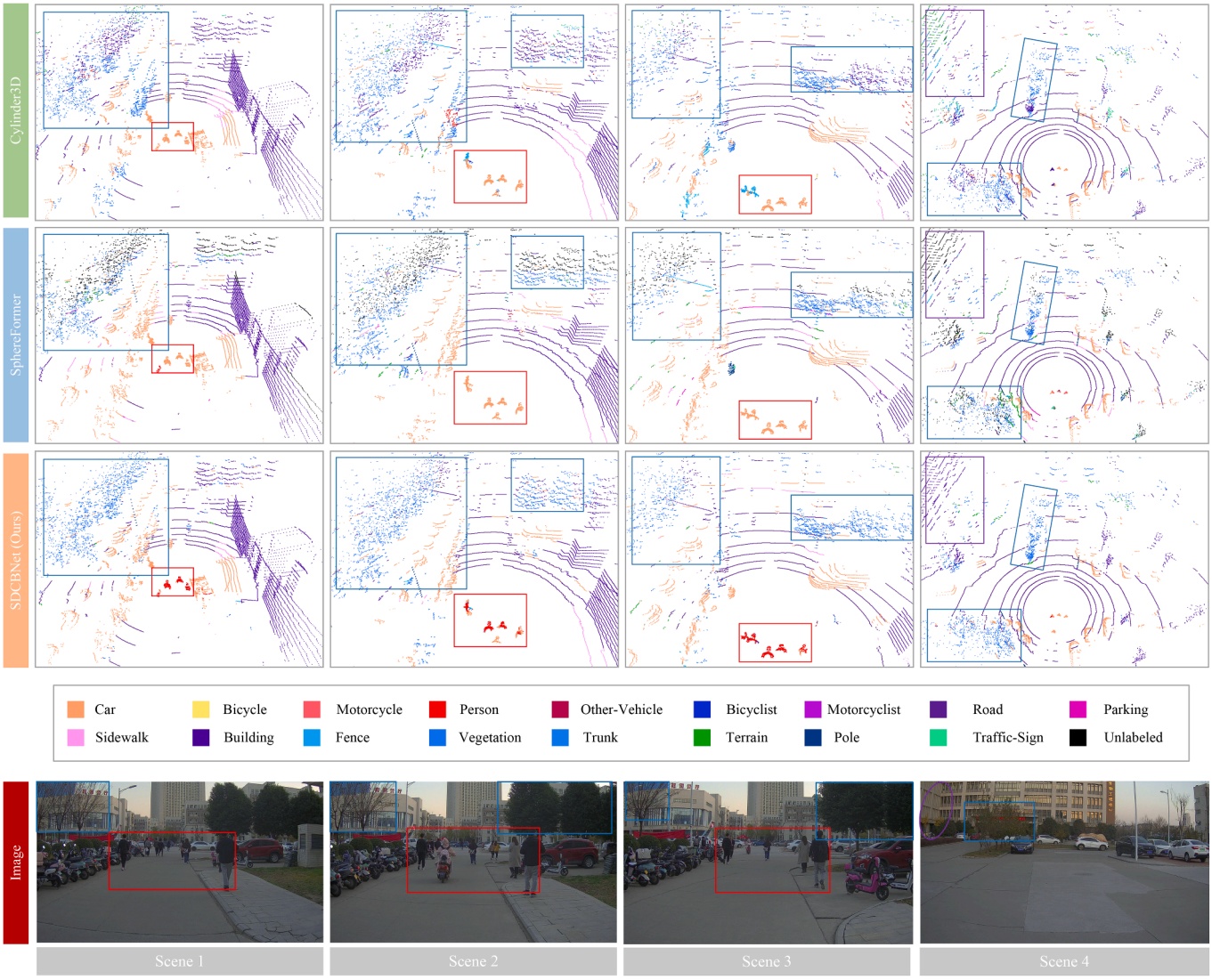


Fig. 5. Qualitative comparison results on our self-collected campus road dataset. The top three rows display the semantic segmentation results of LiDAR point clouds obtained by Cylinder3D, SphereFormer, and our proposed SDCBNet, respectively. The bottom row provides the corresponding RGB images for four typical scenarios (Scene 1 to Scene 4). The colored boxes highlight specific regions where our method demonstrates superior performance.

challenges posed by sparse 16-beam point clouds and complex campus layouts, the proposed network effectively leverages the BEV representation and the CRFS module to capture long-range contextual dependencies. Visually, this comprehensive contextual awareness significantly enhances the segmentation accuracy for large-scale static structures, such as buildings and vegetation, as well as critical dynamic objects like pedestrians. These qualitative results confirm that our approach not only performs well on standardized benchmarks but also effectively adapts to the practical complexities of real-world deployment.

5. Conclusion

This paper presented SDCBNet, a sparse-to-dense Cylinder-BEV network for LiDAR semantic segmentation. The proposed method addresses the limited horizontal context modeling of cylindrical representations, where object features may become fragmented across neighboring cylinder boundaries. By introducing a BEV stream and multi-stage Cylinder-to-BEV feature interaction, SDCBNet aggregates cross-cylinder contextual information and complements fragmented local features while preserving their geometric details. The SDFA module combines max-based and attention-based aggregation to retain salient responses and capture semantically consistent cues, while CRFS selectively injects reliable cross-view information according to the current semantic state. Experimental results on SemanticKITTI and nuScenes demonstrate the effectiveness and generalization ability of SDCBNet. In particular, it achieves an mIoU of 69.5% on the SemanticKITTI validation set, improving the Cylinder3D baseline by 3.4%, with notable gains on small and sparsely sampled foreground categories.

Despite these improvements, the dual-stream architecture and multi-stage feature injection introduce additional computational and memory overhead, which may limit deployment on resource-constrained autonomous driving platforms. Moreover, extremely sparse small objects at long distances remain challenging because limited LiDAR observations provide insufficient geometric and semantic evidence. Future work will explore more lightweight cross-view interaction mechanisms and efficient long-range context modeling to improve deployment efficiency and distant-object perception.

CRediT authorship contribution statement

Liuxin Yu: Writing – original draft, Visualization, Validation, Investigation; **Jun Cheng:** Writing – review & editing, Supervision, Resources, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code, pre-trained models, and the custom dataset used in this study are publicly available at https://github.com/ICV-WTU/code_SDCBNet.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.neunet.2026.109758.

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